

BIOLOGY FOR COMPUTER ENGINEERS

Simple & Easy Study Notes

Module 2 + Module 3 — Questions 21 to Module 3 Q22

MODULE 2

Q21. What are Vaccines? What are the different types?

A **vaccine** is a medicine that teaches your immune system to fight a disease — without actually giving you that disease. It contains a harmless piece of a germ (or instructions to make one). Your body learns to recognize it and builds a defense. If the real germ enters later, your body destroys it quickly.

Types of Vaccines

Type	What it contains	Example
Inactivated (Killed)	Dead germs — safe, may need boosters	Polio vaccine
Live Attenuated (Weakened)	Weakened live germs — strong, long immunity	MMR (Measles, Mumps, Rubella)
DNA Vaccines	DNA plasmid with germ instructions	Under development
mRNA Vaccines	mRNA tells your cells to make a germ protein	COVID-19 vaccines
Subunit Vaccines	Only a piece (protein) of the germ	Hepatitis B vaccine

■ *Memory trick: Vaccines = practice run for your immune system!*

Q22. How does an mRNA Vaccine work? (Step-by-step)

- **Step 1 — Injection:** The mRNA vaccine is injected into your arm muscle.
- **Step 2 — mRNA enters cells:** The mRNA (messenger RNA) enters your muscle cells. It carries instructions to make a specific protein (e.g., the COVID spike protein).
- **Step 3 — Protein is made:** Your cell reads the mRNA and builds the viral protein. This protein is harmless — it just looks like part of a germ.
- **Step 4 — Immune system reacts:** Your immune system spots this strange protein and sends T-cells and B-cells to attack it.
- **Step 5 — Antibodies formed:** B-cells make antibodies against the protein. Memory cells are also created.
- **Step 6 — Future protection:** If the real virus enters your body, your immune system recognizes it immediately and destroys it before you get sick.
- **Bonus — mRNA disappears:** The mRNA breaks down in a few days and is gone. It never enters the cell nucleus or changes your DNA.

■ *Key point: mRNA vaccines do NOT contain the virus. They only carry instructions.*

Q23. What are DNA Vaccines? How do they work + Applications

A **DNA vaccine** uses a small circular piece of DNA called a **plasmid**. This plasmid contains the gene for making a specific antigen (germ protein). When injected, your cells read this DNA and produce the protein, which then triggers immunity.

How it works (simple steps):

- Plasmid DNA is injected into muscle or skin cells.
- The DNA enters the cell nucleus and is copied into mRNA.
- mRNA moves to the cytoplasm and makes the antigen protein.
- The immune system recognizes the antigen and produces antibodies + memory cells.

Applications:

- Infectious diseases (COVID-19, HIV, Influenza, Hepatitis)
- Cancer immunotherapy — target cancer cell proteins
- Veterinary medicine — rabies, foot-and-mouth disease in animals
- Pandemic preparedness — quick design during outbreaks
- Gene therapy research

O24. Explain the working of COVID-19 mRNA Vaccine

Scientists identified that the COVID-19 virus has a **spike protein** on its surface. They made mRNA instructions to produce this spike protein. When injected:

- mRNA enters your muscle cells.
- Cells make the spike protein (harmless by itself).
- Immune system attacks the spike protein, makes antibodies and memory cells.
- If the real COVID virus enters later, antibodies bind to its spike protein and stop it from infecting cells.

■ *The mRNA breaks down after a few days. It never touches your DNA!*

O25. What are the Limitations of mRNA Vaccines?

- **Cold storage needed:** mRNA is fragile — breaks easily. Needs very low temperature (even -70°C). Hard to store/transport in remote areas.
- **Short-lived mRNA:** mRNA degrades quickly, so booster doses may be needed for long-term immunity.
- **Side effects:** Fever, fatigue, arm pain are common. Rare cases of allergic reactions or myocarditis.
- **Delivery problem:** mRNA must be wrapped in lipid nanoparticles to enter cells. Improper delivery = less effective.
- **Expensive technology:** Requires advanced labs and infrastructure — costly compared to traditional vaccines.
- **Disease-specific:** Each disease needs a new mRNA vaccine design, testing, and approval.

O26. What is Whey Protein? Production + Applications

Whey protein is a high-quality protein that comes as a by-product when making cheese from milk. It contains all 9 essential amino acids, making it a **complete protein** — easily digestible and quickly absorbed.

How is it produced?

- Milk is treated with rennet or acid → separates into solid curd (for cheese) and liquid whey.
- The liquid whey is filtered (ultrafiltration/microfiltration) to concentrate the protein.
- Fats and lactose are removed → creates Whey Concentrate, Isolate, or Hydrolysate.
- Dried into powder form for packaging.

Applications:

- Sports/Bodybuilding — muscle growth and recovery
- Medical use — treating malnutrition, surgery recovery

- Food industry — added to bakery, beverages, infant formula
- Weight management — increases feeling of fullness
- Pharma — used in drug formulations

O27. What are Meat Analogs? Types. Production. Limitations

Meat analogs = food that looks, tastes, and feels like meat but is made from plants, fungi, or lab-grown cells. Also called plant-based meat or meat substitutes.

Types:

- **Plant-based:** Made from soy, pea, wheat. E.g., Tofu, Tempeh, TVP.
- **Mycoprotein:** Made from fungi. E.g., Quorn — fibrous texture like meat.
- **Cultured (Lab-grown) meat:** Real animal cells grown in a lab — closest to actual meat.
- **Algae-based:** Made from microalgae — rich in protein and vitamins.

Production steps:

- Protein extraction from soy/pea/wheat
- Texturization — extrusion cooking to create fibrous, meat-like texture
- Formulation — add fats, flavors, colors
- Shaping into burgers, nuggets, sausages
- Packaging under controlled conditions

Limitations:

- Taste/texture not exactly like real meat
- May lack Vitamin B12, iron unless fortified
- Highly processed
- Expensive (especially lab-grown)
- Soy/gluten may cause allergies
- Cultural resistance from consumers

O28. Nutritional Comparison: Meat vs Meat Analogs

Feature	Meat	Meat Analogs
Protein	Complete amino acid profile	High, often supplemented
Fat	High saturated animal fat	Lower fat, from plant oils
Cholesterol	Present — risk if excess	Absent (plant/microbial)
Fiber	Absent	Present (plant-based)
Source	Animals (poultry, fish)	Soy, pea, fungi, algae

O29. What is Biodiesel? Production. Advantages. Limitations

Biodiesel is a clean, renewable fuel made from vegetable oils, animal fats, or used cooking oil. It can be used in diesel engines with little or no changes.

How is it made? (Transesterification)

The reaction: **Triglyceride + Alcohol → Biodiesel (fatty acid esters) + Glycerol**

- Feedstock prep — filter oils, remove impurities

- Mix with methanol + NaOH catalyst → sodium methoxide
- Heat to 50-60°C and stir — triglycerides react to form biodiesel + glycerol
- Settle in layers — biodiesel on top, glycerol at bottom
- Wash with water, dry → clean biodiesel ready!

Advantages:

- Renewable source
- Eco-friendly — less CO, sulfur, particulate emissions
- Biodegradable and non-toxic
- Reduces greenhouse gases
- Works in existing diesel engines

Limitations:

- More expensive than regular diesel
- Slightly lower energy content
- Gels in cold weather
- Shorter shelf life
- Needs large amounts of raw material — competes with food

O30. Production of Detergents from Lipids + Applications

Detergents are **surfactants** — they have a water-loving head (hydrophilic) and an oil-loving tail (hydrophobic). This lets them remove dirt and grease.

Production steps:

- Hydrolysis — lipids (fats/oils) are broken into fatty acids + glycerol
- Hydrogenation — fatty acids become fatty alcohols
- Sulfonation — fatty alcohol + SO₃ → alkyl sulfate (detergent molecule)
- Neutralization with NaOH → water-soluble detergent salt
- Formulation — add builders, fragrances, enzymes → powder or liquid

How cleaning works:

Detergent molecules surround grease particles, forming **micelles**. The oily tail grabs the grease; the watery head pulls it into water. The grease is washed away!

Applications:

- Household cleaning — clothes, utensils
- Industrial — textiles, food processing
- Personal care — shampoos, soaps, toothpaste
- Medical — cleaning hospital equipment
- Agriculture — wetting agents in pesticides

O31. What is a Biosensor? Types + How Glucometer works

A **biosensor** detects biological substances and converts them into an electrical signal. It has 2 parts: a **bioreceptor** (recognizes the target molecule) and a **transducer** (converts the reaction to electrical signal).

Types of Biosensors:

- Electrochemical — measures current/voltage (e.g., Glucometer)

- Optical — detects light changes (e.g., Fiber optic sensor)
- Piezoelectric — measures mass/pressure changes
- Thermal — detects heat from reactions
- Immunosensors — uses antigen-antibody interaction

Glucometer — Step by step:

- Step 1: Blood drop on test strip → touches glucose oxidase enzyme
- Step 2: Reaction: $\text{Glucose} + \text{O}_2 \rightarrow \text{Gluconolactone} + \text{H}_2\text{O}_2$
- Step 3: H_2O_2 reaches graphene-platinum electrode → oxidizes → releases electrons
- Step 4: Electrons generate electric current
- Step 5: More glucose = more current = higher reading on display

■ *Key idea: The amount of current is directly proportional to the blood glucose level!*

O32. Enzymes in Food Processing — Role. Working. Types

Enzymes are biological catalysts (proteins) that speed up reactions without being used up. In food processing, they improve texture, flavor, nutritional value, shelf life, and efficiency.

Key roles:

- Improve texture and consistency
- Enhance flavor and aroma
- Increase nutritional value
- Extend shelf life
- Clarify juices and beverages

How they work:

Enzyme + Substrate → Enzyme-Substrate Complex → Product + Enzyme (reused)

Important Enzymes:

Enzyme	Job	Used in
Amylase	Breaks starch into sugars	Baking, brewing
Protease	Breaks proteins	Meat tenderization, cheese
Lipase	Breaks fats	Dairy flavor development
Lactase	Converts lactose to glucose+galactose	Lactose-free milk
Pectinase	Breaks pectin	Fruit juice extraction
Rennet	Coagulates milk	Cheese production
Invertase	Sucrose → glucose + fructose	Confectionery

O33. Enzymes in Detergent Production

Enzymes in detergents act like tiny scissors — they cut stain molecules into small water-soluble pieces that wash away easily. They work at low temperatures, saving energy!

Why use enzymes in detergents?

- Break down tough stains at low temperatures
- Eco-friendly — less harsh chemicals needed
- Save energy (cold water washing works)

- Protect fabric quality

Enzyme	Removes	How
Protease	Blood, egg, sweat	Breaks proteins into amino acids
Amylase	Starch stains (food, sauces)	Converts starch to simple sugars
Lipase	Oil and grease	Breaks fats into fatty acids + glycerol
Cellulase	Fabric fibers	Removes microfibrils, improves softness
Pectinase	Fruit stains	Breaks down pectin
Mannanase	Food thickener stains	Removes processed food residues

O34. Enzymes in Textile Processing

Enzymes are used as eco-friendly substitutes for harsh chemicals in fabric processing. They work under mild conditions and improve fabric quality.

- **De-sizing:** Amylase removes starch from yarn — improves absorbency
- **Bio-polishing:** Cellulase removes surface fuzz from cotton — gives smooth, shiny finish
- **Denim Finishing (Bio-stoning):** Cellulase gives denim a faded, worn look — replaces stone-washing
- **Bio-bleaching:** Laccase/Peroxidase removes natural pigments — eco-friendly bleaching
- **Degumming (Silk):** Protease removes sericin gum from raw silk — makes it soft and lustrous
- **Color Fading:** Cellulase + Laccase achieve uniform color removal from fabrics

Advantages:

Eco-friendly • Work under mild conditions • Improve quality • Less harsh chemicals

Limitations:

Sensitive to temperature/pH • Higher cost • Need controlled conditions

MODULE 3 — Anatomical Principles for Bioengineering Design

M3-O1. What is Bioengineering? Real-World Applications

Bioengineering = Applying biology + engineering + medicine to solve real-world problems. It creates devices and systems that interact with living organisms.

Applications:

- Healthcare — MRI scanners, pacemakers, artificial limbs, dialysis machines, tissue engineering
- Genetic Engineering — Gene editing, recombinant DNA, insulin production, vaccines
- Agriculture — GM crops resistant to pests and diseases
- Environment — Wastewater treatment, bioremediation, biofuels
- Bio-inspired systems — Sonar from bat echolocation, robotic limbs from human movement

M3-O2. Structure and Working of the Human Brain

The brain is the body's control center. It receives sensory inputs, processes information, and sends responses to muscles and glands through nerves.

Main parts and their functions:

- **Cerebrum (largest part):** Thinking, memory, intelligence, emotions, voluntary movement. Divided into: Frontal (decisions), Parietal (senses), Temporal (hearing/memory), Occipital (vision) lobes.
- **Thalamus:** Relay station — routes sensory signals to correct brain regions.
- **Hypothalamus:** Controls body temperature, hunger, thirst, hormones — homeostasis regulator.
- **Cerebellum:** Balance and coordination — makes movements smooth and precise.
- **Brainstem (midbrain + pons + medulla):** Controls heartbeat, breathing, blood pressure. Pons bridges brain sections; midbrain handles vision/hearing reflexes.

M3-O3. Brain as a CPU — Mapping

The brain works like a CPU — it takes input (senses), processes it, and gives output (actions). But the brain is far more advanced!

Feature	CPU (Computer)	Brain
Architecture	CPU + Memory are separate	Neurons do both processing AND memory
Processing	Mostly sequential	Massively parallel (millions of neurons at once)
Energy	High power consumption	Very energy-efficient
Adaptability	Fixed — needs reprogramming	Learns and reorganizes itself (neuroplasticity)
Speed	Very fast (GHz)	Slower signals but parallel = effective

M3-O4. Brain vs Computer — Differentiation Table

Aspect	Brain	Computer
Structure	Neurons + synapses	Transistors + circuits
Processing	Parallel (millions at once)	Sequential (limited parallel)
Learning	Learns from experience	Needs programming
Speed	Slower signals	Very fast electrical signals
Energy	Very efficient (low power)	High power consumption
Memory	Distributed (all regions)	Centralized (RAM/storage)
Fault Tolerance	Works even if some neurons die	May crash if component fails
Emotion	Has emotions	No emotions — pure logic
Creativity	Can think creatively	Only follows instructions
Self-repair	Limited self-repair	Cannot repair itself

M3-O5. EEG (Electroencephalogram) — Brain Activity + Applications

EEG is a test that records the electrical activity of the brain using electrodes placed on the scalp. It measures brain waves — patterns that tell us what state the brain is in.

Brain Waves:

Wave	Frequency	Mental State
Delta (δ)	0.5 – 4 Hz	Deep sleep, unconscious
Theta (θ)	4 – 8 Hz	Drowsy, meditation, light sleep
Alpha (α)	8 – 13 Hz	Relaxed, calm (eyes closed)

Beta (β)	13 – 30 Hz	Active thinking, concentration
Gamma (γ)	30 – 100 Hz	High cognition, memory, learning

Applications:

- Detecting epilepsy (seizure patterns)
- Sleep disorder diagnosis
- Brain tumor/damage detection
- Monitoring anesthesia depth during surgery
- Brain-Computer Interface (BCI) technology

■ *Advantage: Non-invasive, safe, real-time. Limitation: Low spatial resolution — can't pinpoint exact location.*

M3-O6. Anatomy of the Human Eye

The eye works like a camera — it captures light, focuses it, and converts it to electrical signals sent to the brain.

- **Cornea:** Transparent front layer — first refracts (bends) incoming light
- **Iris:** Colored ring — controls how much light enters by adjusting pupil size
- **Pupil:** The dark opening light passes through
- **Lens:** Flexible structure — focuses light on retina. Changes shape for near/far objects (accommodation)
- **Vitreous humor:** Gel-like substance behind lens — maintains eye shape
- **Retina:** Light-sensitive layer at back — has rod and cone cells that convert light to electrical signals
- **Fovea:** Center of retina — clearest, most detailed vision
- **Optic nerve:** Sends electrical signals from retina to the brain
- **Sclera:** Tough outer white layer — protects and maintains eye shape

M3-O7. Role of Rod and Cone Cells of the Retina

Rod Cells:

- Cylindrical shape
- Work in dim/low light (night vision = scotopic vision)
- Give black-and-white vision — no color detection
- Located mainly in peripheral retina
- More numerous than cones

Cone Cells:

- Conical shape
- Work in bright light (photopic vision)
- Responsible for color and sharp detailed vision
- Concentrated in fovea (center of retina)
- 3 types: sensitive to Red, Green, Blue light

■ *Both rod and cone cells contain photopigments that react to light and generate electrical impulses → optic nerve → brain → image!*

M3-O8. Human Eye vs Camera — Comparison

Part/Function	Human Eye	Camera
Outer cover	Cornea	Lens cover

Light control	Iris (adjusts pupil)	Diaphragm (adjusts aperture)
Opening	Pupil	Aperture
Focusing	Lens (changes shape)	Lens (moves/adjusts)
Image capture	Retina (rods + cones)	Film/Digital sensor
Signal transfer	Optic nerve	Electrical circuits
Image processing	Brain	Image processor/software
Adaptability	Automatic	Manual/programmed
Energy	Biological (nutrients)	Battery

M3-O9. Bionic Eye — What it is and How it Works

A **bionic eye** (retinal implant) is an artificial vision device that restores partial vision in blind people by replacing damaged photoreceptors with electronics.

How it works:

- Miniature camera on special glasses captures the scene
- External processor converts image into digital signals
- Signals sent wirelessly to implanted receiver in/behind the eye
- Receiver sends signals to electrode array implanted on retina
- Electrode array stimulates healthy remaining retinal cells electrically
- Retinal cells send signals via optic nerve to brain
- Brain interprets signals as rough shapes, light, and movement

■ *Result: Not perfect vision, but enough to detect light, shapes, and movement — improves independence!*

M3-O10. Camera Vision System — How it Works

A camera vision system captures, processes, and analyzes visual information for automated decisions.

- **Lens:** Collects and focuses light from object onto sensor
- **Sensor (CCD/CMOS):** Converts light into electrical/digital data
- **Processor:** Applies noise reduction, color correction, enhancement
- **Feature Extraction:** Identifies edges, shapes, objects, patterns in image
- **Output:** Displays result or makes automated decisions (e.g., object detection)

M3-O11. Anatomy of the Human Heart

The heart is a muscular pump with 4 chambers that maintains blood circulation throughout the body, delivering oxygen and removing CO₂.

Double Circulation (Blood flow path):

- Deoxygenated blood → Right Atrium → Tricuspid valve → Right Ventricle
- Right Ventricle → Pulmonary Artery → Lungs (gets oxygenated)
- Oxygenated blood → Pulmonary Vein → Left Atrium → Mitral valve → Left Ventricle
- Left Ventricle (thick walls) → Aorta → Entire Body

■ *Systole = heart contracts (pumps blood). Diastole = heart relaxes (fills with blood).*

M3-O12. Working of a Pacemaker

A **pacemaker** is an electronic implant that regulates the heartbeat when the heart's natural pacemaker (SA node) fails. It has a **pulse generator** (battery + circuits) and **leads** (wires to the heart).

How it works:

- Pacemaker monitors heart's electrical activity continuously
- If heartbeat is too slow or missing → sends electrical impulse through leads
- Impulse stimulates heart muscle to contract → maintains normal beat
- Modern pacemakers are demand-based — only fires when needed
- Rate-responsive — increases rate during exercise, decreases during rest

Types:

- **Single-chamber:** 1 lead in right atrium or ventricle
- **Dual-chamber:** 2 leads — coordinates atrium + ventricle
- **Biventricular (CRT):** Leads in both ventricles — for heart failure patients

M3-O13. Total Artificial Heart (TAH) — Working

A **TAH** completely replaces both ventricles when the natural heart fails. It maintains blood circulation while waiting for a heart transplant.

- Damaged ventricles are surgically removed; atria and major vessels kept
- Artificial device connected to aorta and pulmonary artery
- Right pump: receives deoxygenated blood → sends to lungs
- Left pump: receives oxygenated blood from lungs → sends to body
- External controller regulates pumping rate and pressure
- Powered via driveline connected outside the body

M3-O14. Anatomy and Working of the Lungs

Lungs are the organs of gas exchange — they bring oxygen into the blood and remove CO₂. Located in the chest, protected by ribs.

Structure (airway path): Larynx → Trachea → Primary Bronchi → Secondary Bronchi → Bronchioles → Alveoli

Gas Exchange at Alveoli:

- Alveoli are tiny air sacs surrounded by capillaries
- Oxygen from inhaled air diffuses into blood
- CO₂ from blood diffuses into alveoli
- Oxygenated blood goes to heart → pumped to body

Breathing mechanism:

- **Inhalation:** Diaphragm contracts + rib cage expands → air drawn in
- **Exhalation:** Diaphragm relaxes + rib cage contracts → CO₂ air pushed out

M3-O15. Lungs as a Purification System

The lungs continuously **purify the blood** by removing CO₂ (waste) and adding O₂ (fresh oxygen). They also filter airborne particles.

- Deoxygenated blood (rich in CO₂) enters lungs via pulmonary artery
- CO₂ from blood diffuses into alveoli (lower concentration there)

- O₂ from inhaled air diffuses into blood (higher concentration in air)
- O₂ binds to hemoglobin in red blood cells → carried to all tissues
- CO₂ is expelled during exhalation
- Extra: Nasal passages, mucus, and cilia trap dust and microorganisms before air reaches alveoli

■ *The lungs filter ~10,000 liters of air every day!*

M3-O16. Ventilator — Working and Benefits

A **ventilator** (mechanical ventilator) helps patients breathe when they cannot do so on their own. It mechanically pushes oxygen-rich air into the lungs.

How it works:

- Oxygen-rich air is delivered through endotracheal tube into lungs
- Ventilator pushes air under controlled pressure → alveoli expand
- Gas exchange occurs: O₂ into blood, CO₂ into alveoli
- Ventilator allows exhalation → CO₂-rich air exits
- Cycle repeats continuously — all parameters (pressure, frequency, O₂ level) controlled

Used for:

- ICU patients with respiratory failure
- COVID-19, pneumonia, COPD, asthma attacks
- Surgery under anesthesia
- Trauma, coma, cardiac arrest
- Neuromuscular disorders (weak breathing muscles)

M3-O17. Working of Artificial Lungs (TAL)

A **Total Artificial Lung (TAL)** fully replaces natural lungs for gas exchange using a device called an **oxygenator**. Used when lungs are severely damaged or removed.

- Deoxygenated blood is diverted from the body into the TAL device
- Blood flows through a semipermeable membrane (oxygenator)
- O₂ is supplied to the other side of membrane → diffuses into blood
- CO₂ from blood diffuses out into device
- Oxygenated blood pumped back into the body
- Pumps, sensors, and controllers regulate blood flow and O₂ levels

■ *Used in ARDS, severe infections, and bilateral pneumonectomy (both lungs removed) as bridge to transplant.*

M3-O18. Kidney as Filtration System — Working

The kidney filters blood using millions of tiny units called **nephrons**. It removes waste (urea, creatinine) and keeps useful substances.

Step-by-step process in the nephron:

- **Glomerulus / Bowman's Capsule:** Blood enters at high pressure → filters water, salts, glucose, urea into filtrate. Large proteins and blood cells stay in blood.
- **Proximal Convoluted Tubule (PCT):** Most glucose, amino acids, and water are reabsorbed back into blood.
- **Loop of Henle:** More water and salts reabsorbed; concentrates the filtrate.

- **Distal Convolted Tubule (DCT):** Fine regulation — secretes excess ions and wastes; adjusts pH.
- **Collecting Duct:** Final water reabsorption → remaining fluid = urine → flows to ureter → bladder.

M3-O19. Blood and Urine Flow in the Kidney

Blood flow path:

Renal Artery → Afferent Arteriole → Glomerulus (filtration) → Efferent Arteriole → Peritubular Capillaries (reabsorption) → Renal Vein → Exit kidney

Urine formation path:

Bowman's Capsule → PCT → Loop of Henle → DCT → Collecting Duct → Renal Pelvis → Ureter → Urinary Bladder → Excreted

M3-O20. Dialysis — Use and Working

Dialysis = artificial kidney process. Used when kidneys fail to clean the blood. Works on diffusion through a semipermeable membrane.

Hemodialysis — How it works:

- Blood withdrawn from patient's body through a tube
- Blood passes through dialyzer (artificial kidney) — contains semipermeable membrane
- Dialysis fluid (dialysate) flows on the other side of membrane
- Waste products (urea, salts, toxins) diffuse from blood into dialysate
- Excess water is removed by filtration
- Purified blood returned to patient
- Process takes 3–5 hours, done 3× per week typically

When is dialysis needed?

- Chronic kidney disease
- Acute kidney failure
- Poisoning or toxic substance accumulation
- Severe infection affecting kidneys
- Bridge until kidney transplant

M3-O21. Various Functions of the Kidney

The kidney does much more than just make urine! Its main functions:

- **Blood filtration:** Removes urea, creatinine, excess salts — keeps blood clean
- **Electrolyte balance:** Regulates sodium, potassium, chloride levels
- **Blood pressure regulation:** Produces renin hormone → controls blood volume and vessel constriction
- **RBC production support:** Secretes erythropoietin → stimulates bone marrow to make red blood cells
- **Vitamin D activation:** Converts inactive Vitamin D to active form → needed for calcium absorption and bone health
- **Acid-base balance:** Regulates blood pH by excreting acids and conserving bicarbonate
- **Homeostasis:** Overall fluid and electrolyte balance maintenance

M3-O22. Working of an Artificial Kidney

An **artificial kidney** is a bioengineered device that replicates nephron functions — filtration and reabsorption — using advanced technology.

How it works:

- Blood enters through inlet tube
- Stage 1 — Hemofilter: semipermeable silicon nanotechnology membrane filters out waste, salts, and excess water (ultrafiltration)
- Stage 2 — Cell Bioreactor: living kidney cells selectively reabsorb glucose, salts, and water back into blood
- Waste and excess water go to waste outlet → bladder → excreted as urine
- Purified blood returned to body via outlet tube

■ *Advantage over dialysis: Continuous operation + biological reabsorption = closer to a real kidney. Can be implantable!*